

Exploiting Out-of-Band Information for Millimeter-Wave MIMO Channel Estimation: Performance in Static and Dynamic Scenarios

F. Pasic, M. Mussbah, S. Schwarz, M. Rupp and C. F. Mecklenbräuker

EuCNC & 6G Summit, June 2-5, 2026 - Malaga, Spain



Technische
Universität Wien



- future wireless communication systems
 - support wide range of data-intensive applications
 - increasing demand for high data rates



- future wireless communication systems
 - support wide range of data-intensive applications
 - increasing demand for high data rates
- sub-6 GHz frequency bands
 - limited available bandwidth
 - unable to meet future data rate requirements

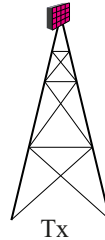


- future wireless communication systems
 - support wide range of data-intensive applications
 - increasing demand for high data rates
- sub-6 GHz frequency bands
 - limited available bandwidth
 - unable to meet future data rate requirements
- millimeter-wave (mmWave) frequency bands
 - larger bandwidths available
 - enable multi-Gbps data rates

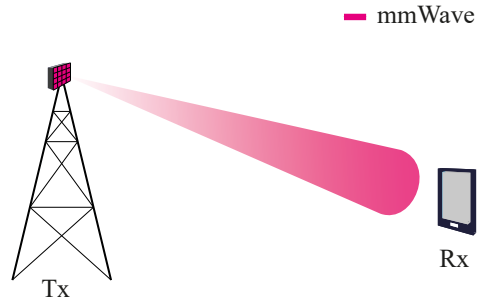


- mmWave systems utilize antenna arrays

— mmWave



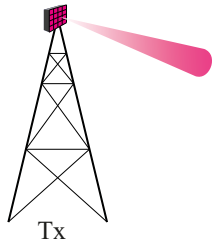
- mmWave systems utilize antenna arrays
 - apply MIMO techniques



MIMO ... multiple-input multiple-output

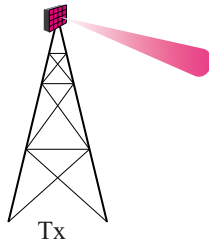
- mmWave systems utilize antenna arrays
 - apply MIMO techniques
 - low pre-beamforming SNR

— mmWave



MIMO ... multiple-input multiple-output

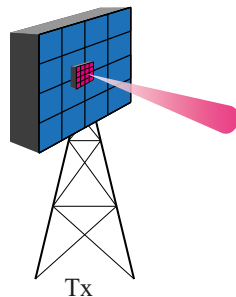
- mmWave systems utilize antenna arrays
 - apply MIMO techniques
 - low pre-beamforming SNR
 - **problem**: challenging link establishment



— mmWave

MIMO ... multiple-input multiple-output

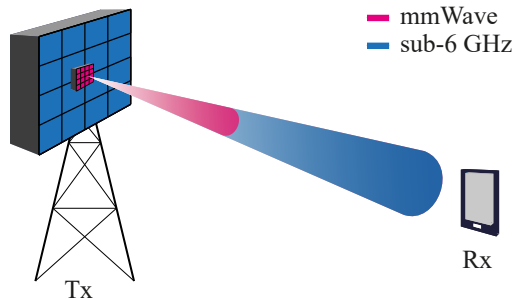
- mmWave systems utilize antenna arrays
 - apply MIMO techniques
 - low pre-beamforming SNR
 - **problem:** challenging link establishment
- mmWave deployment in conjunction with sub-6 GHz systems
 - sub-6 GHz → favorable propagation conditions



— mmWave
— sub-6 GHz

MIMO ... multiple-input multiple-output

- mmWave systems utilize antenna arrays
 - apply MIMO techniques
 - low pre-beamforming SNR
 - **problem**: challenging link establishment
- mmWave deployment in conjunction with sub-6 GHz systems
 - sub-6 GHz → favorable propagation conditions
 - ▷ **use out-of-band channel information to improve mmWave link establishment**



MIMO ... multiple-input multiple-output

- **out-of-band aided channel estimation method for mmWave MIMO communication systems**
 - estimate mmWave channel with the aid sub-6 GHz out-of-band information

- **out-of-band aided channel estimation method for mmWave MIMO communication systems**
 - estimate mmWave channel with the aid sub-6 GHz out-of-band information
- **demonstrate performance through numerical simulations in terms of achievable spectral efficiency (SE)**
 - static and dynamic environments
 - various pilot configurations
 - wide range of receiver velocities
 - varying propagation conditions (LOS and NLOS)

System Model

Simulation Results

Summary and Conclusion

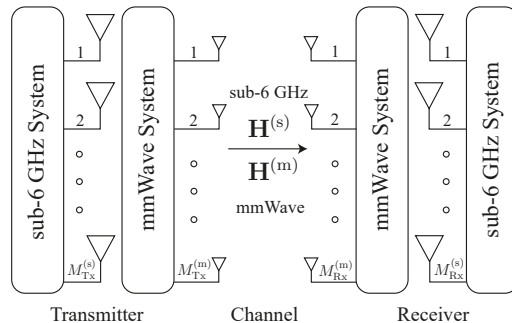
System Model

Simulation Results

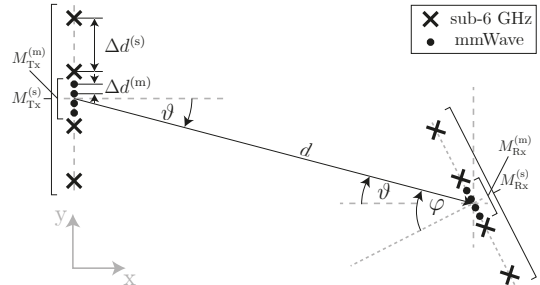
Summary and Conclusion

- multi-band MIMO OFDM system

OFDM ... orthogonal freq.-division multiplexing



- multi-band MIMO OFDM system
 - co-located and aligned ULAs

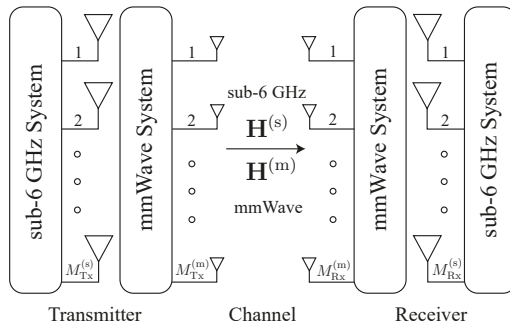


OFDM ... orthogonal freq.-division multiplexing

ULA ... uniform linear array

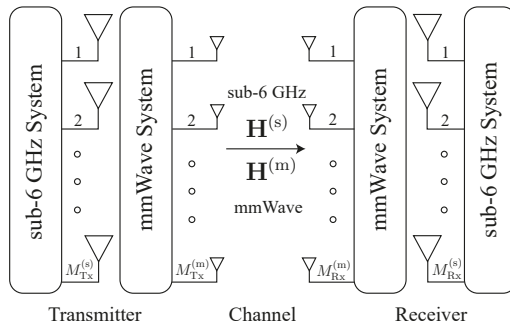
- multi-band MIMO OFDM system
 - co-located and aligned ULAs
 - fully digital beamforming

OFDM ... orthogonal freq.-division multiplexing
ULA ... uniform linear array



- multi-band MIMO OFDM system
 - co-located and aligned ULAs
 - fully digital beamforming
 - TDD system

OFDM ... orthogonal freq.-division multiplexing
ULA ... uniform linear array
TDD ... time-division duplex



- multi-band MIMO OFDM system
 - co-located and aligned ULAs
 - fully digital beamforming
 - TDD system
 - Ricean channel model

$$\mathbf{H}^{(s)} = \underbrace{\sqrt{\eta^{(s)}}}_{\text{pathloss}} \left(\underbrace{\sqrt{\frac{\kappa^{(s)}}{1 + \kappa^{(s)}}} \mathbf{H}_{\text{fs}}^{(s)}}_{\text{free-space}} + \underbrace{\sqrt{\frac{1}{1 + \kappa^{(s)}}} \mathbf{H}_{\text{sp}}^{(s)}}_{\text{stochastic part}} \right)$$
$$\mathbf{H}^{(m)} = \underbrace{\sqrt{\eta^{(m)}}}_{\text{pathloss}} \left(\underbrace{\sqrt{\frac{\kappa^{(m)}}{1 + \kappa^{(m)}}} \mathbf{H}_{\text{fs}}^{(m)}}_{\text{free-space}} + \underbrace{\sqrt{\frac{1}{1 + \kappa^{(m)}}} \mathbf{H}_{\text{sp}}^{(m)}}_{\text{stochastic part}} \right)$$

OFDM ... orthogonal freq.-division multiplexing

ULA ... uniform linear array

TDD ... time-division duplex

- training phase
 - both bands used: sub-6 GHz and mmWave

- data transmission phase

$(\cdot)^{(s)}$... sub-6 GHz band

$(\cdot)^{(m)}$... mmWave band

LOS ... line-of-sight

- training phase
 - both bands used: sub-6 GHz and mmWave
 - 1st phase $\implies$ non-beamformed channel estimates $\tilde{\mathbf{H}}^{(s)}$ and $\tilde{\mathbf{H}}^{(m)}$

- data transmission phase

$(\cdot)^{(s)}$... sub-6 GHz band

$(\cdot)^{(m)}$... mmWave band

LOS ... line-of-sight

- training phase
 - both bands used: sub-6 GHz and mmWave
 - 1st phase $\implies$ non-beamformed channel estimates $\tilde{\mathbf{H}}^{(s)}$ and $\tilde{\mathbf{H}}^{(m)}$
 - 2nd phase $\implies$ beamformed (LOS) channel estimate $\hat{\mathbf{H}}^{(m)}$
- data transmission phase

$(\cdot)^{(s)}$... sub-6 GHz band

$(\cdot)^{(m)}$... mmWave band

LOS ... line-of-sight

- training phase
 - both bands used: sub-6 GHz and mmWave
 - 1st phase $\implies$ non-beamformed channel estimates $\tilde{\mathbf{H}}^{(s)}$ and $\tilde{\mathbf{H}}^{(m)}$
 - 2nd phase $\implies$ beamformed (LOS) channel estimate $\hat{\mathbf{H}}^{(m)}$
 - resulting channel estimate $\bar{\mathbf{H}}^{(m)} \implies$ calculate precoder $\bar{\mathbf{F}}^{(m)}$ and combiner $\bar{\mathbf{Q}}^{(m)}$
- data transmission phase

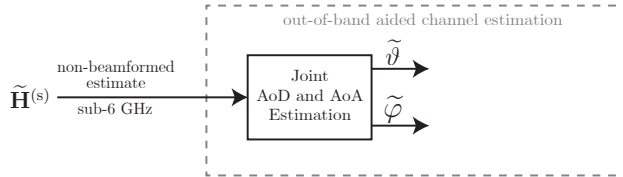
$(\cdot)^{(s)}$... sub-6 GHz band

$(\cdot)^{(m)}$... mmWave band

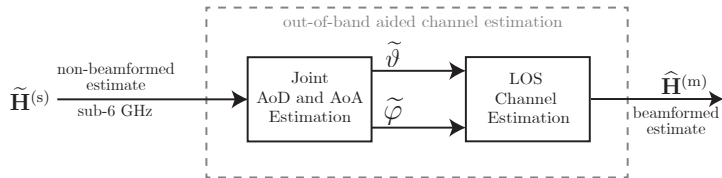
LOS ... line-of-sight



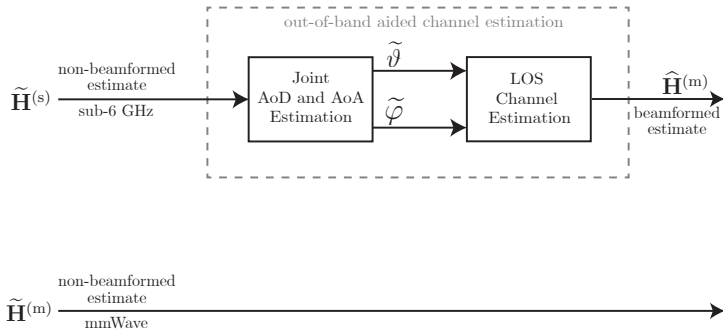
F. Pasic, M. Hofer, M. Mussbah, S. Sangodoyin, S. Caban, S. Schwarz, T. Zemen, M. Rupp, A. F. Molisch and C. F. Mecklenbräuer, "Millimeter wave MIMO channel estimation using sub-6 GHz out-of-band information," *IEEE Transactions on Communications*, July 2025



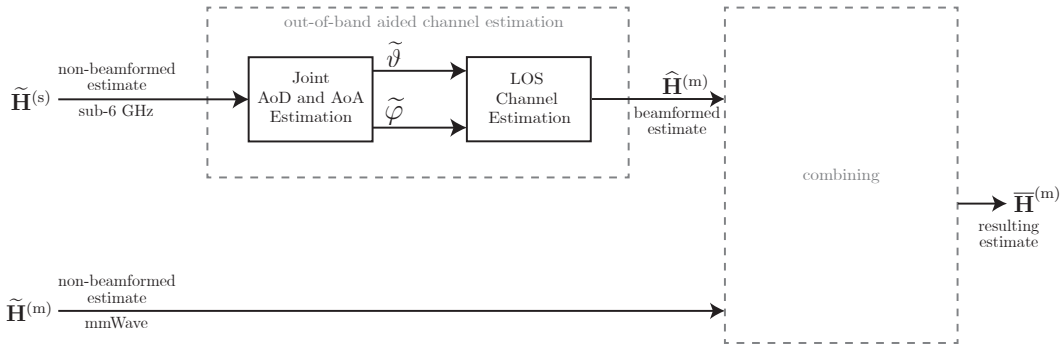
F. Pasic, M. Hofer, M. Mussbah, S. Sangodoyin, S. Caban, S. Schwarz, T. Zemen, M. Rupp, A. F. Molisch and C. F. Mecklenbräuer, "Millimeter wave MIMO channel estimation using sub-6 GHz out-of-band information," *IEEE Transactions on Communications*, July 2025



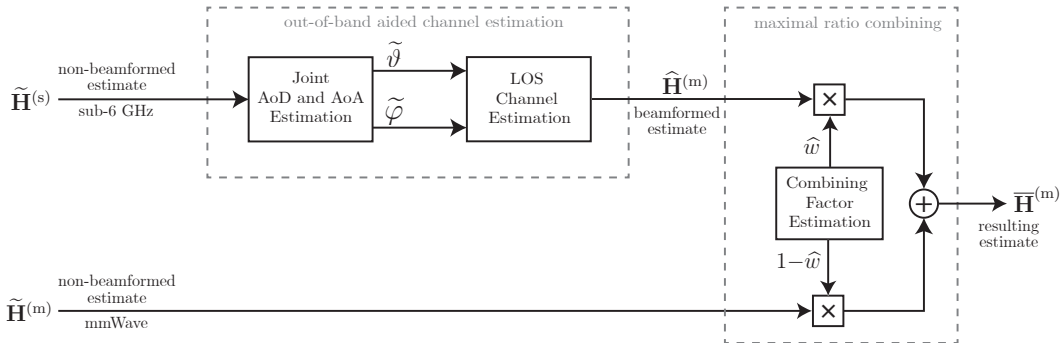
F. Pasic, M. Hofer, M. Mussbah, S. Sangodoyin, S. Caban, S. Schwarz, T. Zemen, M. Rupp, A. F. Molisch and C. F. Mecklenbräuer, "Millimeter wave MIMO channel estimation using sub-6 GHz out-of-band information," *IEEE Transactions on Communications*, July 2025



F. Pasic, M. Hofer, M. Mussbah, S. Sangodoyin, S. Caban, S. Schwarz, T. Zemen, M. Rupp, A. F. Molisch and C. F. Mecklenbräuer, "Millimeter wave MIMO channel estimation using sub-6 GHz out-of-band information," *IEEE Transactions on Communications*, July 2025



F. Pasic, M. Hofer, M. Mussbah, S. Sangodoyin, S. Caban, S. Schwarz, T. Zemen, M. Rupp, A. F. Molisch and C. F. Mecklenbräuer, "Millimeter wave MIMO channel estimation using sub-6 GHz out-of-band information," *IEEE Transactions on Communications*, July 2025



F. Pasic, M. Hofer, M. Mussbah, S. Sangodoyin, S. Caban, S. Schwarz, T. Zemen, M. Rupp, A. F. Molisch and C. F. Mecklenbräuer, "Millimeter wave MIMO channel estimation using sub-6 GHz out-of-band information," *IEEE Transactions on Communications*, July 2025

- training phase
 - both bands used: sub-6 GHz and mmWave
 - 1st phase $\implies$ non-beamformed channel estimates $\tilde{\mathbf{H}}^{(s)}$ and $\tilde{\mathbf{H}}^{(m)}$
 - 2nd phase $\implies$ beamformed (LOS) channel estimate $\hat{\mathbf{H}}^{(m)}$
 - resulting channel estimate $\bar{\mathbf{H}}^{(m)} \implies$ calculate precoder $\bar{\mathbf{F}}^{(m)}$ and combiner $\bar{\mathbf{Q}}^{(m)}$
- data transmission phase
 - used mmWave band only
 - transmit data using precoder $\bar{\mathbf{F}}^{(m)}$ and combiner $\bar{\mathbf{Q}}^{(m)}$

$(\cdot)^{(s)}$... sub-6 GHz band

$(\cdot)^{(m)}$... mmWave band

LOS ... line-of-sight

System Model

Simulation Results

Summary and Conclusion

- investigate achievable spectral efficiency

Parameter	Value	
Frequency Band	sub-6 GHz	mmWave
Carrier Frequency f_c	2.55 GHz	25.5 GHz
Wavelength λ	11.76 cm	1.176 cm
Bandwidth B	20.16 MHz	403.2 MHz
Subcarrier Spacing Δf	60 kHz	120 kHz
Number of Data Symbols K_D	N/A	7
Transmit Power P_T	30 dBm	30 dBm
Antenna Configuration ($M_{Rx} \times M_{Tx}$)	8×8	8×8
RMS Delay Spread σ_τ	1148 ns	841 ns
Number of Realizations L_r	1000	1000

- investigate achievable spectral efficiency
- three channel estimation methods
 - perfect CSI
 - conventional LS estimation
 - out-of-band aided MRC (OOBA-MRC)

Parameter	Value	
Frequency Band	sub-6 GHz	mmWave
Carrier Frequency f_c	2.55 GHz	25.5 GHz
Wavelength λ	11.76 cm	1.176 cm
Bandwidth B	20.16 MHz	403.2 MHz
Subcarrier Spacing Δf	60 kHz	120 kHz
Number of Data Symbols K_D	N/A	7
Transmit Power P_T	30 dBm	30 dBm
Antenna Configuration ($M_{Rx} \times M_{Tx}$)	8×8	8×8
RMS Delay Spread σ_τ	1148 ns	841 ns
Number of Realizations L_r	1000	1000

CSI ... channel state information

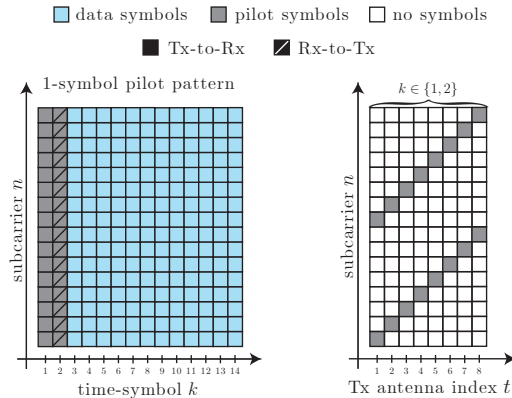
- investigate achievable spectral efficiency
- three channel estimation methods
 - perfect CSI
 - conventional LS estimation
 - out-of-band aided MRC (OOBA-MRC)
- three pilot schemes

Parameter	Value	
Frequency Band	sub-6 GHz	mmWave
Carrier Frequency f_c	2.55 GHz	25.5 GHz
Wavelength λ	11.76 cm	1.176 cm
Bandwidth B	20.16 MHz	403.2 MHz
Subcarrier Spacing Δf	60 kHz	120 kHz
Number of Data Symbols K_D	N/A	7
Transmit Power P_T	30 dBm	30 dBm
Antenna Configuration ($M_{Rx} \times M_{Tx}$)	8×8	8×8
RMS Delay Spread σ_τ	1148 ns	841 ns
Number of Realizations L_r	1000	1000

CSI ... channel state information

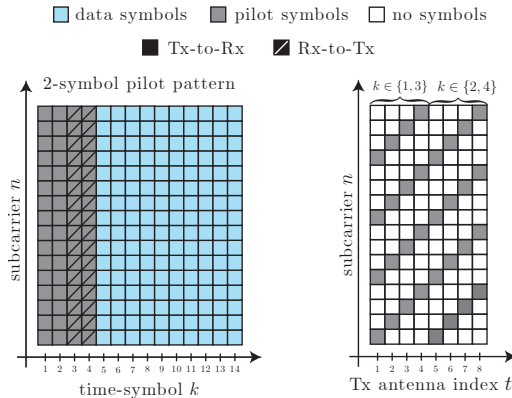
- investigate achievable spectral efficiency
- three channel estimation methods
 - perfect CSI
 - conventional LS estimation
 - out-of-band aided MRC (OOBA-MRC)
- three pilot schemes
 - 1-pilot pattern ($K_P = 1$)

CSI ... channel state information



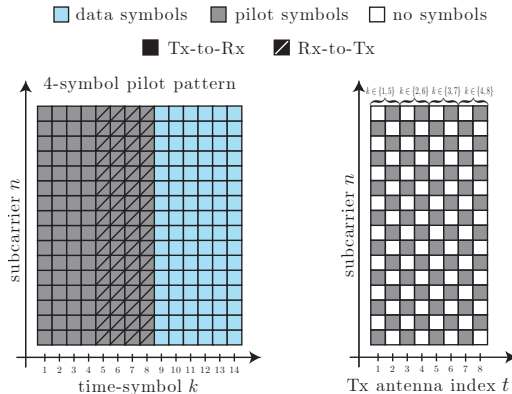
- investigate achievable spectral efficiency
- three channel estimation methods
 - perfect CSI
 - conventional LS estimation
 - out-of-band aided MRC (OOBA-MRC)
- three pilot schemes
 - 1-pilot pattern ($K_P = 1$)
 - 2-pilot pattern ($K_P = 2$)

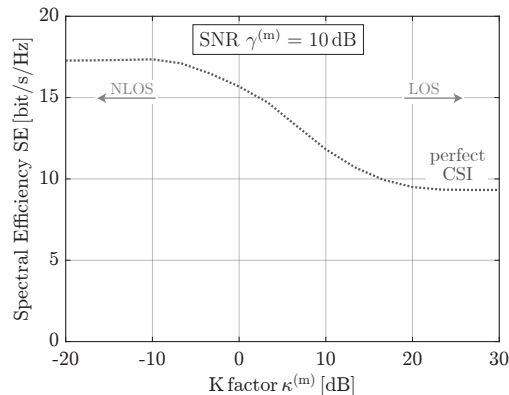
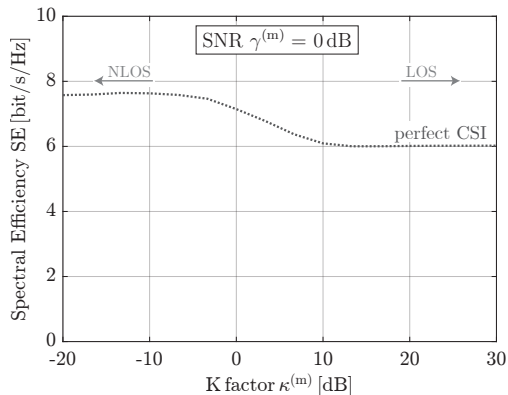
CSI ... channel state information

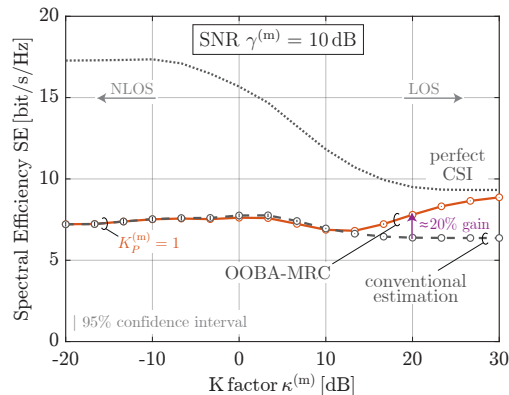
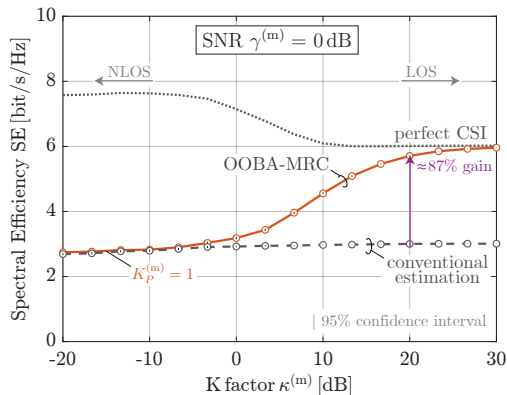


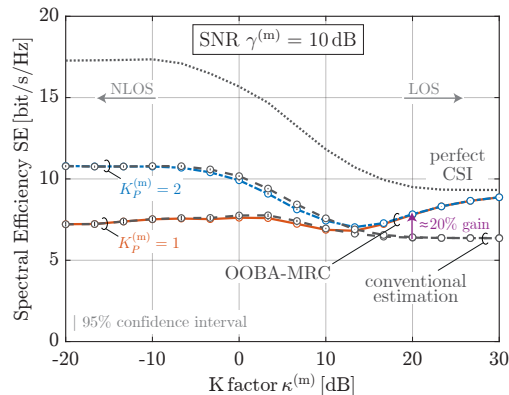
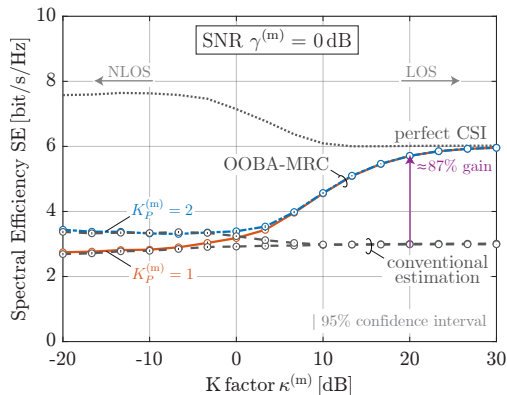
- investigate achievable spectral efficiency
- three channel estimation methods
 - perfect CSI
 - conventional LS estimation
 - out-of-band aided MRC (OOBA-MRC)
- three pilot schemes
 - 1-pilot pattern ($K_P = 1$)
 - 2-pilot pattern ($K_P = 2$)
 - 4-pilot pattern ($K_P = 4$)

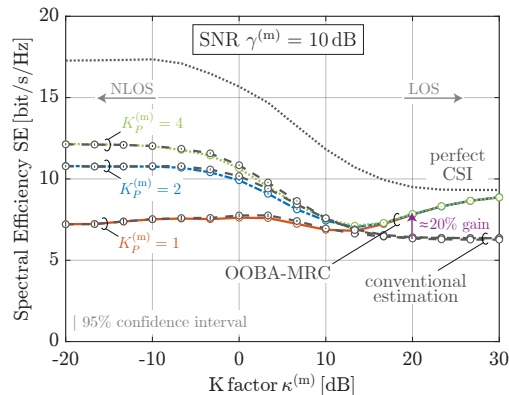
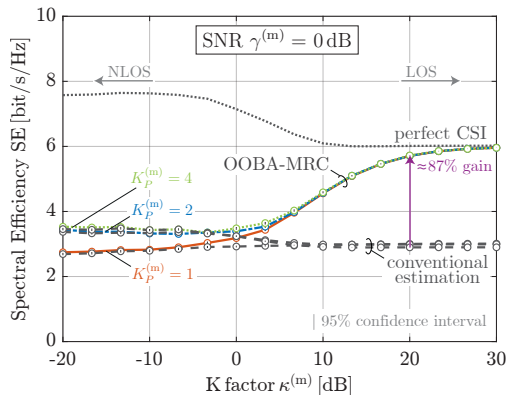
CSI ... channel state information

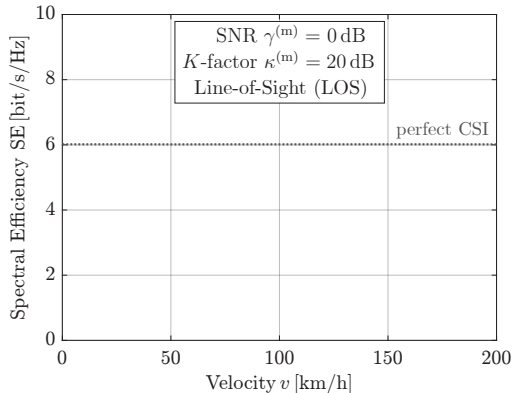
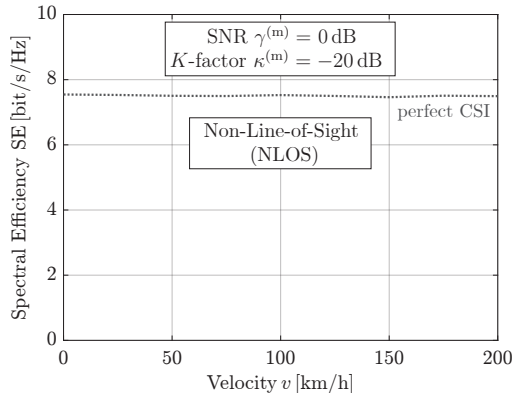


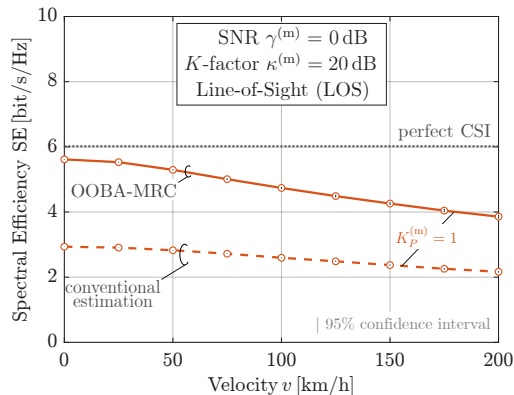
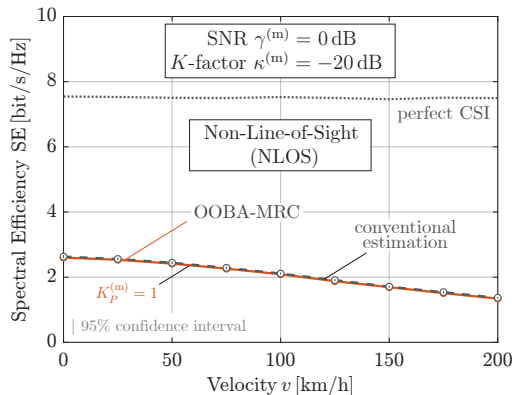


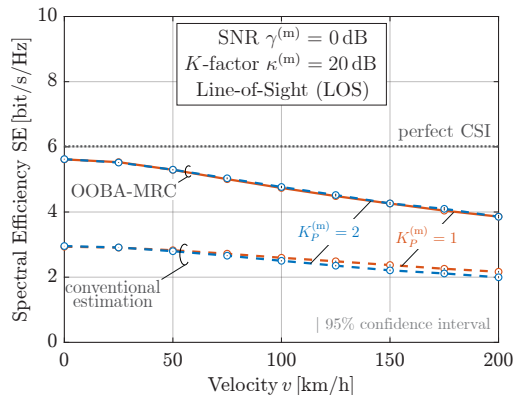
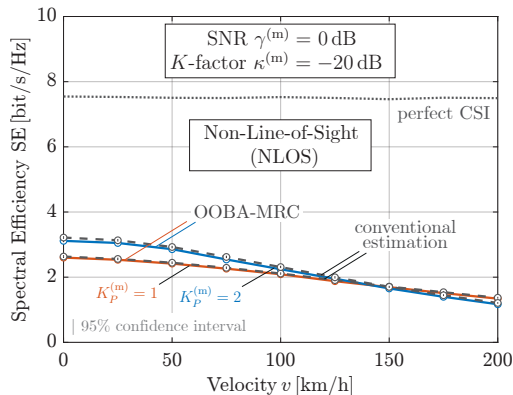


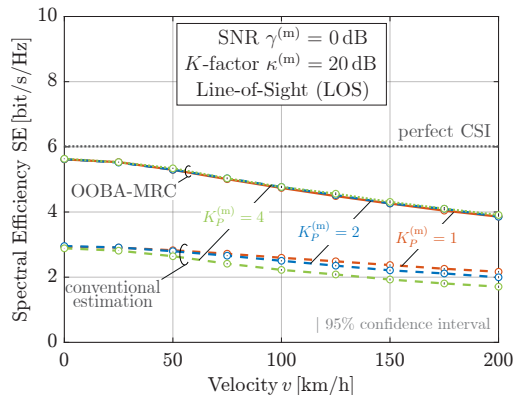
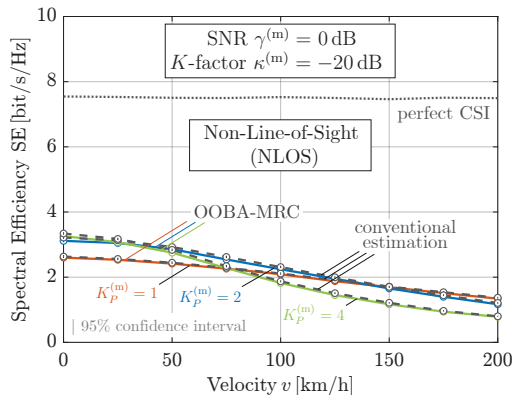












System Model

Simulation Results

Summary and Conclusion

- **exploiting sub-6 GHz out-of-band information within channel estimation improves performance**
 - achievable gains strongly influenced by SNR, Rician K -factor and receiver velocity
 - higher performance gains achieved in LOS scenarios compared to NLOS scenarios

- **exploiting sub-6 GHz out-of-band information within channel estimation improves performance**
 - achievable gains strongly influenced by SNR, Rician K -factor and receiver velocity
 - higher performance gains achieved in LOS scenarios compared to NLOS scenarios
- **static scenarios: channel aging negligible**
 - increasing number of pilot symbols → improves or preserves SE performance

- **exploiting sub-6 GHz out-of-band information within channel estimation improves performance**
 - achievable gains strongly influenced by SNR, Rician K -factor and receiver velocity
 - higher performance gains achieved in LOS scenarios compared to NLOS scenarios
- **static scenarios: channel aging negligible**
 - increasing number of pilot symbols → improves or preserves SE performance
- **dynamic scenarios: channel aging becomes a dominant limiting factor**
 - channel changes significantly during estimation interval → older estimates lose relevance
 - increasing number of pilot symbols → may decrease in SE due to outdated channel information

Thank you for your attention!
Questions?

Faruk Pasic
faruk.pasic@tuwien.ac.at

- each transmit antenna is assigned pilot symbols $\phi^{(b)}[n, k]$ that are known at the receiver

$$\phi_t^{(b)}[n, k] = \begin{cases} \phi^{(b)}[n, k], & \text{if } (n, k, t) \in \mathcal{P}^{(b)} \\ 0, & \text{else} \end{cases} \quad (1)$$

- where the set of pilot positions $\mathcal{P}^{(b)}$ is defined by

$$\begin{aligned} \mathcal{P}^{(b)} = \{ & (n, k, t) : n \in \left\{ t_m, t_m + \frac{M_{\text{Tx}}^{(b)}}{K_{\text{P}}^{(b)}}, \dots, N^{(b)} - \frac{M_{\text{Tx}}^{(b)}}{K_{\text{P}}^{(b)}} + t_m \right\}, \\ & k \in \{1, 2, \dots, 2K_{\text{P}}^{(b)}\}, \quad t \in \{1, 2, \dots, M_{\text{Tx}}^{(b)}\}, \\ & t_m = \left(t - 1 \bmod \frac{M_{\text{Tx}}^{(b)}}{K_{\text{P}}^{(b)}} \right) + 1 \} \}. \end{aligned} \quad (2)$$

- input-output relationship

$$\mathbf{y}^{(b)}[n, k] = \mathbf{H}^{(b)}[n, k]\phi^{(b)}[n, k] + \mathbf{w}^{(b)}[n, k], \quad (3)$$

$\phi^{(b)}[n, k] \dots$ pilot symbols

$\mathbf{y}^{(b)}[n, k] \dots$ received symbols

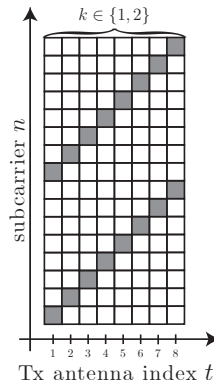
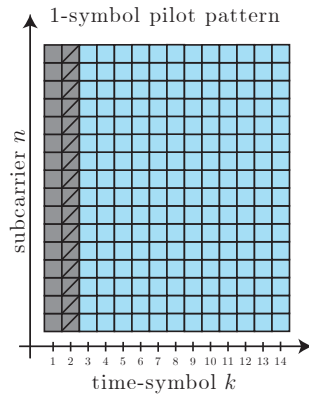
$\mathbf{w}^{(b)}[n, k] \sim \mathcal{CN}\left(0, \sigma_{w^{(b)}}^2 \mathbf{I}_{M_{\text{Rx}}^{(b)}}\right) \dots$ AWGN

$\sigma_{w^{(b)}}^2 \dots$ noise variance

- least-square (LS) estimation at pilot positions
- linear interpolation estimates the channel at non-occupied subcarrier positions
- result: channel estimate $\tilde{\mathbf{H}}^{(b)}[n] \in \mathbb{C}^{M_{\text{Rx}}^{(b)} \times M_{\text{Tx}}^{(b)}}$

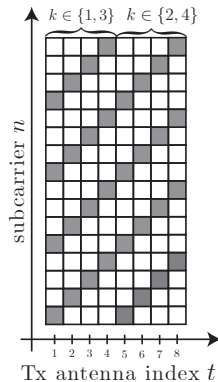
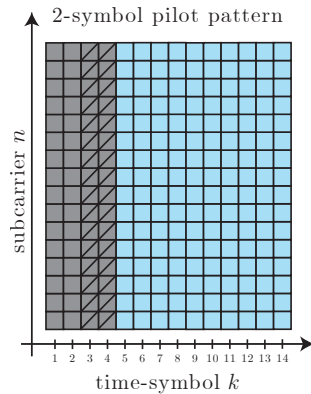
□ data symbols ■ pilot symbols □ no symbols

■ Tx-to-Rx ▤ Rx-to-Tx



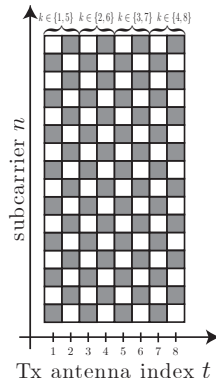
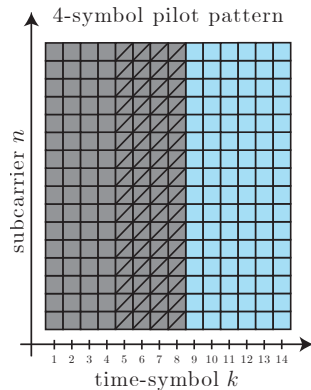
■ data symbols ■ pilot symbols □ no symbols

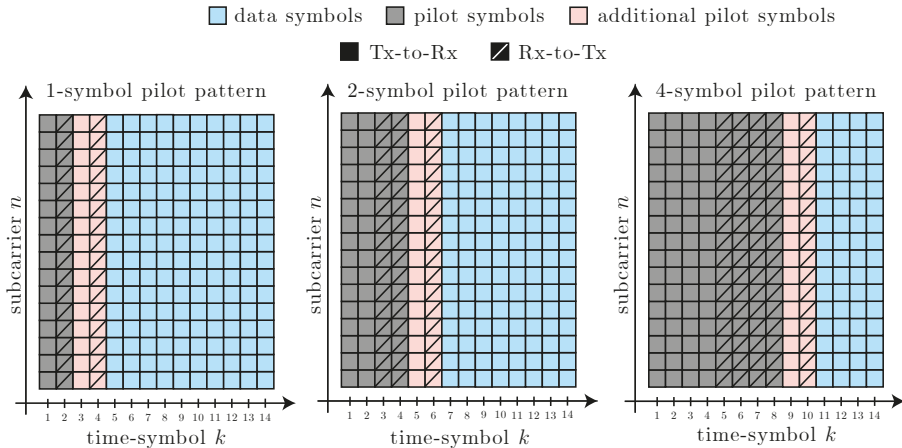
■ Tx-to-Rx ▤ Rx-to-Tx



data symbols
 pilot symbols
 no symbols

Tx-to-Rx
 Rx-to-Tx





- use resulting mmWave channel estimate $\overline{\mathbf{H}}^{(m)}[n]$ to calculate the precoding and combining

$$\overline{\mathbf{H}}^{(m)}[n] = \overline{\mathbf{Q}}^{(m)}[n] \overline{\mathbf{\Sigma}}^{(m)}[n] \left(\overline{\mathbf{F}}^{(m)}[n] \right)^H \quad (4)$$

$\overline{\mathbf{F}}^{(m)}[n] \in \mathbb{C}^{M_{\text{Tx}}^{(m)} \times \ell_{\text{max}}} \dots$ precoding matrix, $\overline{\mathbf{Q}}^{(m)}[n] \in \mathbb{C}^{M_{\text{Rx}}^{(m)} \times \ell_{\text{max}}} \dots$ combining matrix,

$\overline{\mathbf{\Sigma}}^{(m)}[n] \in \mathbb{C}^{\ell_{\text{max}} \times \ell_{\text{max}}} \dots$ singular value matrix

$\ell_{\text{max}} = \min \left(M_{\text{Rx}}^{(m)}, M_{\text{Tx}}^{(m)} \right) \dots$ maximum number of streams

- total transmit power constraint satisfied by precoder for all subcarriers

$$\left\| \overline{\mathbf{F}}^{(m)}[n] \left(\overline{\mathbf{P}}^{(m)}[n] \right)^{1/2} \right\|_F^2 = P_{\text{T}}^{(m)} \quad (5)$$

$\overline{\mathbf{P}}^{(m)}[n] \in \mathbb{C}^{\ell_{\text{max}} \times \ell_{\text{max}}} \dots$ power loading matrix, $P_{\text{T}}^{(m)} \dots$ total transmit power

- input-output relationship for data transmission phase

$$\mathbf{y}^{(m)}[n, k] = \left(\overline{\mathbf{Q}}^{(m)}[n] \right)^H \mathbf{H}^{(m)}[n, k] \overline{\mathbf{F}}^{(m)}[n] \left(\overline{\mathbf{P}}^{(m)}[n] \right)^{1/2} \mathbf{x}^{(m)}[n, k] + \left(\overline{\mathbf{Q}}^{(m)}[n] \right)^H \mathbf{w}^{(m)}[n, k] \quad (6)$$

$\mathbf{x}^{(m)}[n, k] \in \mathbb{C}^{\ell_{\max} \times 1} \dots$ transmit symbols

$\mathbf{y}^{(m)}[n, k] \in \mathbb{C}^{\ell_{\max} \times 1} \dots$ received symbols

$\overline{\mathbf{F}}^{(m)}[n] \in \mathbb{C}^{M_{\text{Tx}}^{(m)} \times \ell_{\max}} \dots$ precoding matrix

$\overline{\mathbf{Q}}^{(m)}[n] \in \mathbb{C}^{M_{\text{Rx}}^{(m)} \times \ell_{\max}} \dots$ combining matrix,

$\overline{\mathbf{P}}^{(m)}[n] \in \mathbb{C}^{\ell_{\max} \times \ell_{\max}} \dots$ power loading matrix

$\mathbf{w}^{(m)}[n, k] \sim \mathcal{CN} \left(0, \sigma_{w^{(m)}}^2 \mathbf{I}_{M_{\text{Rx}}^{(m)}} \right) \dots$ AWGN

- achievable spectral efficiency in bits/s/Hz averaged over $N^{(m)}$ subcarriers and $K_D^{(m)}$ time-symbols

$$SE = \frac{1}{N^{(m)} K_D^{(m)}} \sum_{n=1}^{N^{(m)}} \sum_{k=K_P^{(m)}+1}^{K^{(m)}} \sum_{\mu=1}^{\ell_{\max}} \log_2 (1 + \text{SINR}_{\mu}[n, k]) \quad (7)$$

with the effective signal-to-interference-and-noise-ratio (SINR) for the stream μ

$$\text{SINR}_{\mu}[n, k] = \frac{|\overline{G}_{\mu, \mu}^{(m)}[n, k]|^2}{\sum_{\substack{\nu=1 \\ \nu \neq \mu}}^{\ell_{\max}} |\overline{G}_{\mu, \nu}^{(m)}[n, k]|^2 + \left(\sigma_{\mu}^2[n, k] + \sigma_{w^{(m)}}^2 \right) \|\overline{\mathbf{Q}}_{:, \mu}^{(m)}[n, k]\|^2} \quad (8)$$

with the entries of the channel gain matrix $\overline{G}_{\mu, \nu}^{(m)}[n, k]$ with $\mu, \nu \in \{1, \dots, \ell_{\max}\}$

and channel gain matrix $\overline{\mathbf{G}}^{(m)}[n, k] \in \mathbb{C}^{\ell_{\max} \times \ell_{\max}}$ for the n -th subcarrier and k -th time-symbol

$$\overline{\mathbf{G}}^{(m)}[n, k] = \left(\overline{\mathbf{Q}}^{(m)}[n] \right)^H \mathbf{H}^{(m)}[n, k] \overline{\mathbf{F}}^{(m)}[n] \left(\overline{\mathbf{P}}^{(m)}[n] \right)^{1/2} \quad (9)$$

- $\sigma_{\mu}^2[n, k]$ diagonal entries (variance) of the estimation error covariance matrix for the n -th subcarrier and k -th time-symbol

$$\overline{\mathbf{C}}_{\varepsilon}^{(m)}[n, k] = \frac{1}{\ell_{\max}} \bar{\varepsilon}^{(m)}[n, k] \left(\bar{\varepsilon}^{(m)}[n, k] \right)^H \quad (10)$$

with the estimation error matrix

$$\bar{\varepsilon}^{(m)}[n, k] = \mathbf{G}^{(m)}[n, k] - \overline{\mathbf{G}}^{(m)}[n, k] \quad (11)$$

where $\bar{\varepsilon}^{(m)}[n, k]$ represents the error between the channel gain matrix with the estimated precoder and combiner, $\overline{\mathbf{G}}^{(m)}[n, k]$, and the channel gain matrix with the perfectly designed precoder and combiner

$$\mathbf{G}^{(m)}[n, k] = \left(\mathbf{Q}^{(m)}[n, k] \right)^H \mathbf{H}^{(m)}[n, k] \mathbf{F}^{(m)}[n, k] \left(\mathbf{P}^{(m)}[n, k] \right)^{1/2} \quad (12)$$

- goal: optimally combine two channel estimates
 - in-band non-beamformed channel estimates $\tilde{\mathbf{H}}^{(s)}[n]$ and $\tilde{\mathbf{H}}^{(m)}[n]$
 - out-of-band aided LOS channel estimate $\hat{\mathbf{H}}^{(m)}[n]$
- why not using mmWave LOS AoD/AoA?
 - significant propagation losses + lack of beamforming at the transmitter
 - low pre-beamforming SNR $\implies$ highly imprecise mmWave channel estimate $\tilde{\mathbf{H}}^{(m)}[n]$
 - sub-6 GHz has higher SNR $\implies$ more accurate estimate $\tilde{\mathbf{H}}^{(s)}[n]$ and accurate AoD/AoA estimation

- jointly estimate the AoD/AoA using 2D-Bartlett beamformer within the sub-6 GHz system

$$\tilde{\mathbf{h}}^{(s)}[n] = \text{vec} \left(\tilde{\mathbf{H}}^{(s)}[n] \right) \in \mathbb{C}^{M_{\text{Tx}}^{(s)} M_{\text{Rx}}^{(s)} \times 1} \quad (13)$$

- estimate the AoA/AoD for LOS component and focus solely on the zero-delay channel tap

$$\tilde{\mathbf{h}}_{\text{LOS}}^{(s)}[\tau] = \tilde{\mathbf{h}}^{(s)}[\tau] \delta[\tau] \quad (14)$$

- estimate covariance matrix

$$\tilde{\mathbf{C}}^{(s)} = \tilde{\mathbf{h}}_{\text{LOS}}^{(s)}[n] \left(\tilde{\mathbf{h}}_{\text{LOS}}^{(s)}[n] \right)^H \quad (15)$$

- compute angular power spectrum of the 2D-Bartlett beamformer

$$P_{\vartheta, \varphi}^{(s)}(\vartheta_{\text{tr}}, \varphi_{\text{tr}}) = \frac{\left(\mathbf{a}^{(s)}(\vartheta_{\text{tr}}, \varphi_{\text{tr}}) \right)^H \tilde{\mathbf{C}}^{(s)} \mathbf{a}^{(s)}(\vartheta_{\text{tr}}, \varphi_{\text{tr}})}{\left(\mathbf{a}^{(s)}(\vartheta_{\text{tr}}, \varphi_{\text{tr}}) \right)^H \mathbf{a}^{(s)}(\vartheta_{\text{tr}}, \varphi_{\text{tr}})}, \quad (16)$$

with joint steering vector

$$\mathbf{a}^{(s)}(\vartheta_{\text{tr}}, \varphi_{\text{tr}}) = \left(\mathbf{a}_{\text{Tx}}^{(s)}(\vartheta_{\text{tr}}) \right)^* \otimes \mathbf{a}_{\text{Rx}}^{(s)}(\varphi_{\text{tr}}) \quad (17)$$

- estimated AoD and AoA found through the maxima of the angular power spectrum

$$\tilde{\vartheta}, \tilde{\varphi} = \arg \max_{\vartheta_{\text{tr}}, \varphi_{\text{tr}}} \left\{ P_{\vartheta, \varphi}^{(s)}(\vartheta_{\text{tr}}, \varphi_{\text{tr}}) \right\}. \quad (18)$$

- additional training phase: estimated AoD and AoA used to precode and combine the pilot symbols in the direction of the LOS component
- input-output relationship for the additional training phase

$$y^{(m)}[n, k] = \left(\mathbf{a}_{\text{Rx}}^{(m)}(\tilde{\varphi}) \right)^H \mathbf{H}^{(m)}[n, k] \mathbf{a}_{\text{Tx}}^{(m)}(\tilde{\vartheta}) \phi^{(m)}[n, k] + \left(\mathbf{a}_{\text{Rx}}^{(m)}(\tilde{\varphi}) \right)^H \mathbf{w}^{(m)}[n, k] \quad (19)$$

- LS estimation and linear interpolation $\implies$ effective beamformed scalar channel estimate $\hat{G}^{(m)}[n]$
- isolate LOS component

$$\hat{G}_{\text{LOS}}^{(m)}[\tau] = \hat{G}^{(m)}[\tau] \delta[\tau] \quad (20)$$

- reconstructed out-of-band aided mmWave MIMO channel estimate for the LOS component

$$\hat{\mathbf{H}}^{(m)}[n] = \mathbf{a}_{\text{Rx}}^{(m)}(\tilde{\varphi}) \hat{G}_{\text{LOS}}^{(m)}[n] \left(\mathbf{a}_{\text{Tx}}^{(m)}(\tilde{\vartheta}) \right)^H \quad (21)$$

- goal: optimally combine two channel estimates
 - in-band non-beamformed channel estimate $\tilde{\mathbf{H}}^{(m)}[n]$
 - out-of-band aided LOS channel estimate $\hat{\mathbf{H}}^{(m)}[n]$
- optimum weighting factor (MRC) for each specific combination of K -factor and SNR

$$\overline{\mathbf{H}}^{(m)}[n] = \hat{w} \hat{\mathbf{H}}^{(m)}[n] + (1 - \hat{w}) \tilde{\mathbf{H}}^{(m)}[n] \quad (22)$$

with

$$\hat{w} \approx \frac{M_{\text{Tx}}^{(m)} M_{\text{Rx}}^{(m)} \sigma_{w^{(m)}}^2}{\frac{M_{\text{Tx}}^{(m)} M_{\text{Rx}}^{(m)}}{1 + c_{\kappa} \tilde{\kappa}^{(s)}} + \left(1 + M_{\text{Tx}}^{(m)} M_{\text{Rx}}^{(m)}\right) \sigma_{w^{(m)}}^2} \quad (23)$$